Artificial Intelligence Prototype for Clinical Diagnosis from Medical Imaging

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Abstract—The interpretation of medical images such as chest X-rays is a critical task that faces challenges of variability, workload, and human error. Although artificial intelligence (AI) has achieved impressive performance in controlled research settings, there remains a gap between high-performing prototypes and clinically deployable systems. This work presents a standardized methodological framework for developing deep-learning-based diagnostic tools, designed to be generalizable across imaging modalities. As a case study, we implemented and validated the framework on the domain of chest X-ray images for pneumonia detection using a ResNet-18 convolutional neural network trained through transfer learning. The model achieved a validation accuracy of 94% and an area under the ROC curve (AUC) of 1.000, while on the held-out test set it reached 94% accuracy, 0.93 macro F1-score, and an AUC of 0.986. These results demonstrate the feasibility of reproducible and interpretable AI systems for medical imaging, supporting the transition from experimental research to clinical applications.

Index Terms—Deep Learning, Medical Imaging, Transfer Learning, Chest X-Ray, Pneumonia Detection, Diagnostic AI, Generalization.

I. Introduction

Medical imaging has become a cornerstone of modern healthcare, enabling the visualization of internal structures and the early detection of pathological changes. Among imaging modalities, chest radiography remains one of the most common and cost-effective diagnostic tools, widely used in screening and evaluating respiratory and cardiac diseases.

However, the manual interpretation of chest X-rays remains subject to human limitations such as visual fatigue, inter-observer variability, and high workload. In low-resource settings, the shortage of trained radiologists further exacerbates diagnostic delays and inaccuracies. These challenges motivate the exploration of artificial intelligence (AI) to support radiological analysis.

Recent advances in deep learning, particularly convolutional neural networks (CNNs), have shown remarkable success in classifying and detecting abnormalities in medical images. Nevertheless, literature reviews and meta-analyses [1]–[3] highlight persistent methodological issues: inconsistent evaluation protocols, lack of external validation, and limited reproducibility. Consequently, many research prototypes fail to generalize to real clinical environments.

This study directly addresses these issues by developing a reproducible and standardized framework for AI-assisted diagnosis in medical imaging. The chest X-ray pneumonia classification task serves as a validation domain, with the ultimate goal of establishing transferable principles applicable to other imaging modalities. Beyond reporting a single model's performance, the focus lies on methodological rigor, interpretability, and potential clinical integration.

II. Methodology and Prototype Development

A. Dataset

We used the Chest X-Ray Images (Pneumonia) dataset published by P. Mooney on Kaggle, which contains over 5,800 chest X-ray images categorized into two classes: Normal and Pneumonia. The dataset was split into training, validation, and test sets following the original structure.

B. Preprocessing

All images were resized to 224×224 pixels and normalized using ImageNet statistics. Data augmentation was applied during training, including random rotations, flips, and brightness variations, to improve generalization.

C. Model Architecture

A ResNet-18 convolutional neural network was used as the base architecture, initialized with ImageNet pretrained weights. The model's final fully connected layer was replaced to output two classes. Transfer learning was applied by fine-tuning all layers using the FastAI library with mixed-precision training on an NVIDIA RTX 3060 Ti GPU.

D. Training Configuration

The model was trained for eight epochs with a batch size of 32 using the Adam optimizer and one-cycle learning rate policy. Training and validation were performed using stratified data loaders to ensure balanced class representation.

E. Evaluation Metrics

Model performance was evaluated using accuracy, precision, recall, F1-score, and the area under the receiver operating characteristic curve (AUC). Interpretability was assessed through confusion matrices and ROC plots for both validation and test datasets.

III. Results and Evaluation

The fine-tuned ResNet-18 achieved high discriminative capability. On the validation subset, it obtained:

- Accuracy: 94%
- AUC: 1.000
- F1-score (macro): 0.94

On the independent test subset:

- Accuracy: 94%
- AUC: 0.986
- F1-score (macro): 0.93

Figure 1 shows the validation confusion matrix, while Figures 2 and 3 present the ROC curves for validation and test sets, respectively. The model shows a strong ability to distinguish between normal and pneumonia cases with minimal misclassifications.

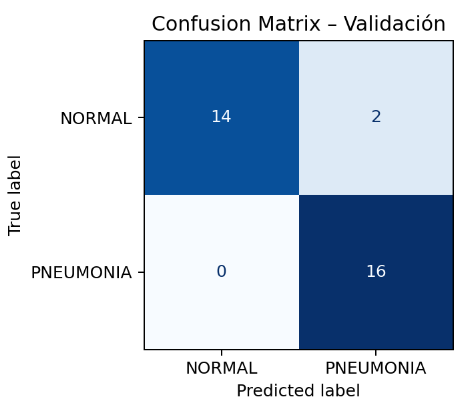

Fig. 1. Confusion Matrix – Validation.

These metrics indicate the robustness of the model and its potential as a reliable diagnostic assistant. Minor confusion occurred primarily between normal and mild pneumonia cases, which is consistent with human diagnostic variability.

IV. Conclusions

This research presented a practical and reproducible framework for developing AI-based diagnostic systems from medical images. Using chest X-rays as a validation domain, the model achieved strong performance metrics while maintaining transparency and simplicity of implementation.

Key findings include:

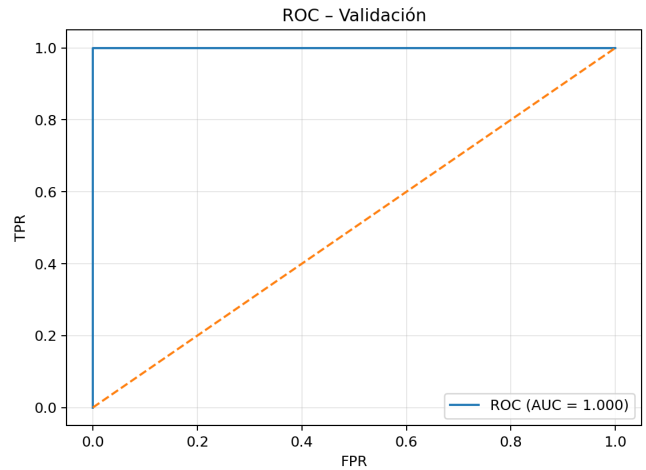

Fig. 2. ROC curve on validation set (AUC = 1.000).

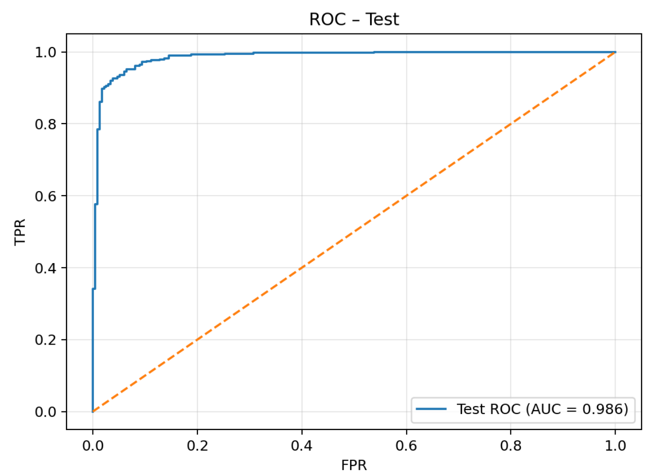

Fig. 3. ROC curve on test set (AUC = 0.986).

- Transfer learning with ResNet-18 provides excellent performance with limited data and computation.
- A standardized pipeline with explicit preprocessing, augmentation, and validation strategies improves reproducibility.
- The framework can be adapted to other medical imaging modalities, facilitating broader clinical applications.

Future work includes extending the framework to multi-class classification problems, incorporating Grad-CAM explainability visualizations, and performing external validation on independent datasets to further assess generalization.

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References

[1] H. P. Chan, R. K. Samala, L. M. Hadjiiski, and C. Zhou, “Deep Learning in Medical Image Analysis,” Advances in Experimental Medicine and Biology, vol. 1213, pp. 3–21, 2020. doi: 10.1007/978-3-030-33128-3_1.

[2] G. K. Thakur, A. Thakur, S. Kulkarni, N. Khan, and S. Khan, “Deep Learning Approaches for Medical Image Analysis and Diagnosis,” Cureus, vol. 16, no. 5, p. e59507, 2024. doi: 10.7759/cureus.59507.

[3] R. Aggarwal, V. Sounderajah, G. Martin, et al., “Diagnostic accuracy of deep learning in medical imaging: a systematic review and meta-analysis,” npj Digital Medicine, vol. 4, no. 65, 2021. doi: 10.1038/s41746-021-00438-z.